

– COORDINATES, VECTOR CALCULUS –

$$\vec{x} = x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}} = r \hat{\mathbf{r}} = \rho \hat{\boldsymbol{\rho}} + z \hat{\mathbf{z}}; \quad d^3x = dx dy dz = r^2 d\Omega dr = r^2 \sin \theta dr d\theta d\phi = \rho d\rho d\phi dz$$

$$d\vec{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin \theta d\phi \hat{\boldsymbol{\phi}} = d\rho \hat{\boldsymbol{\rho}} + \rho d\phi \hat{\boldsymbol{\phi}} + dz \hat{\mathbf{z}}$$

$$\vec{\nabla} f = \partial_x f \hat{\mathbf{x}} + \partial_y f \hat{\mathbf{y}} + \partial_z f \hat{\mathbf{z}} = \partial_r f \hat{\mathbf{r}} + \frac{\partial_\theta f}{r} \hat{\boldsymbol{\theta}} + \frac{\partial_\phi f}{r \sin \theta} \hat{\boldsymbol{\phi}} = \partial_\rho f \hat{\boldsymbol{\rho}} + \frac{\partial_\phi f}{\rho} \hat{\boldsymbol{\phi}} + \partial_z f \hat{\mathbf{z}}$$

$$\vec{\nabla} \cdot \vec{v} = \partial_x v_x + \partial_y v_y + \partial_z v_z = \frac{\partial_r(r^2 v_r)}{r^2} + \frac{\partial_\theta(\sin \theta v_\theta) + \partial_\phi v_\phi}{r \sin \theta} = \frac{\partial_\rho(\rho v_\rho) + \partial_\phi v_\phi}{\rho} + \partial_z v_z$$

$$\begin{aligned} \vec{\nabla} \times \vec{v} &= (\partial_y v_z - \partial_z v_y) \hat{\mathbf{x}} + (\partial_z v_x - \partial_x v_z) \hat{\mathbf{y}} + (\partial_x v_y - \partial_y v_x) \hat{\mathbf{z}} \\ &= \frac{1}{r} \left\{ \frac{1}{\sin \theta} [\partial_\theta(\sin \theta v_\phi) - \partial_\phi v_\theta] \hat{\mathbf{r}} + \left[\frac{\partial_\phi v_r}{\sin \theta} - \partial_r(r v_\theta) \right] \hat{\boldsymbol{\theta}} + [\partial_r(r v_\theta) - \partial_\theta v_r] \hat{\boldsymbol{\phi}} \right\} \\ &= \left(\frac{\partial_\phi v_z}{\rho} - \partial_z v_\phi \right) \hat{\boldsymbol{\rho}} + (\partial_z v_\rho - \partial_\rho v_z) \hat{\boldsymbol{\phi}} + \frac{1}{\rho} [\partial_\rho(\rho v_\phi) - \partial_\phi v_\rho] \hat{\mathbf{z}} \end{aligned}$$

$$\nabla^2 f = (\partial_x^2 + \partial_y^2 + \partial_z^2) f = \frac{1}{r^2} \left\{ \partial_r(r^2 \partial_r f) + \frac{1}{\sin \theta} \left[\partial_\theta(\sin \theta \partial_\theta f) + \frac{\partial_\phi^2 f}{\sin \theta} \right] \right\} = \frac{\partial_\rho(\rho \partial_\rho f)}{\rho} + \frac{\partial_\phi^2 f}{\rho^2} + \partial_z^2 f$$

$$\int_{\vec{a}}^{\vec{b}} \vec{\nabla} f \cdot d\vec{l} = f(\vec{b}) - f(\vec{a}); \quad \int (\vec{\nabla} \times \vec{v}) \cdot \hat{\mathbf{n}} da = \oint \vec{v} \cdot d\vec{l}; \quad \int \vec{\nabla} \cdot \vec{v} d^3x = \oint \vec{v} \cdot \hat{\mathbf{n}} da$$

$$\text{Dirac: } \int f(\vec{x}) \delta(\vec{x} - \vec{a}) d^3x = f(\vec{a}), \quad \int_{-\infty}^x \delta(x - a) dx = \Theta(x - a)$$

$$\int f(x) \delta'(x - a) dx = -f'(a); \quad \delta(f(x)) = \sum_i \frac{\delta(x - x_i)}{|f'(x_i)|}, \quad f(x_i) = 0$$

$$\vec{\nabla} \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) = -\vec{\nabla}' \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) = -\frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3}, \quad \nabla^2 \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) = -4\pi \delta^3(\vec{x} - \vec{x}')$$

$$\text{Green: } \int_V (\vec{\nabla} \phi \cdot \vec{\nabla} \psi + \phi \nabla^2 \psi) d^3x = \oint_S \phi \partial_n \psi da, \quad \int_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) d^3x = \oint_S (\phi \partial_n \psi - \psi \partial_n \phi) da$$

– ELECTROSTATICS –

$$\vec{F}(\vec{x}) = \frac{q q'}{4\pi\epsilon_0} \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} = q \vec{E}(\vec{x}), \quad \frac{1}{4\pi\epsilon_0} \simeq 9.0 \times 10^9 \text{ m/F}; \quad dq \sim \lambda dl \sim \sigma da \sim \rho d^3x$$

$$\text{Open boundaries: } \vec{E}(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int \rho(\vec{x}') \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} d^3x', \quad \Phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3x'$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}, \quad \int \vec{E} \cdot \hat{\mathbf{n}} da = \frac{Q}{\epsilon_0}; \quad \vec{\nabla} \times \vec{E} = \vec{0}, \quad \vec{E} = -\vec{\nabla} \Phi, \quad \int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l} = \Phi(\vec{a}) - \Phi(\vec{b}), \quad \nabla^2 \Phi = -\frac{\rho}{\epsilon_0}$$

$$\text{Interfaces: } [\vec{E}_{out}(\vec{x}) - \vec{E}_{in}(\vec{x})] \cdot \hat{\mathbf{n}}(\vec{x}) = \frac{\sigma(\vec{x})}{\epsilon_0}, \quad \hat{\mathbf{n}}(\vec{x}) \times [\vec{E}_{out}(\vec{x}) - \vec{E}_{in}(\vec{x})] = \vec{0}$$

$$\text{Potential energy: } W = \frac{1}{2} \int \rho \Phi d^3x = \frac{\epsilon_0}{2} \int E^2 d^3x$$

Boundary Value Problems

$$G(\vec{x}, \vec{x}') = \frac{1}{|\vec{x} - \vec{x}'|} + F(\vec{x}, \vec{x}'), \quad \nabla'^2 G(\vec{x}, \vec{x}') = -4\pi\delta(\vec{x} - \vec{x}'), \quad \nabla'^2 F(\vec{x}, \vec{x}') = 0$$

$$G_D(\vec{x}, \vec{x}' \in S) = 0 \rightarrow \Phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int_V \rho(\vec{x}') G_D(\vec{x}, \vec{x}') d^3x' - \frac{1}{4\pi} \oint_S \Phi(\vec{x}') \frac{\partial G_D(\vec{x}, \vec{x}')}{\partial n'} da'$$

$$\frac{\partial G_N}{\partial n'}(\vec{x}, \vec{x}' \in S) = -\frac{4\pi}{S} \rightarrow \Phi(\vec{x}) = \langle \Phi \rangle_S + \frac{1}{4\pi\epsilon_0} \int_V \rho(\vec{x}') G_N(\vec{x}, \vec{x}') d^3x' + \frac{1}{4\pi} \oint_S G_N \frac{\partial \Phi}{\partial n'} da'$$

Separation of variables

$$\text{rectangular: } \Phi(x, y, z) = \sum_{\alpha\beta\gamma} X_\alpha(x) Y_\beta(y) Z_\gamma(z), \quad \frac{\nabla^2 \Phi}{\Phi} = \frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} = 0 = \pm\alpha^2 \pm \beta^2 \pm \gamma^2$$

$$\text{signs, allowed values from bc: } +\alpha^2 \rightarrow e^{\pm\alpha x}, \sinh / \cosh \alpha x; \quad -\beta^2 \rightarrow e^{\pm i\beta y}, \sin / \cos \beta y$$

$$\text{spherical: } \Phi(r, \theta, \phi) = \frac{U(r)}{r} P(\theta) Q(\phi), \quad \frac{(r \sin \theta)^2}{UPQ} \nabla^2 \Phi = 0 = (r \sin \theta)^2 \left[\frac{U''}{U} + \frac{(\sin \theta P')'}{r^2 \sin \theta P} \right] + \frac{Q''}{Q}$$

$$\frac{Q''}{Q} = -m^2 \rightarrow r^2 \frac{U''}{U} = \frac{m^2}{\sin^2 \theta} - \frac{(\sin \theta P')'}{\sin \theta P} = l(l+1) \rightarrow U(r) = Ar^{l+1} + Br^{-l}, P(\theta) = P_l^m(\cos \theta)$$

$$Q(\phi) = e^{\pm im\phi}, \sin / \cos m\phi; \quad \Phi(\phi = 2\pi) = \Phi(\phi = 0) \rightarrow \text{integer } m; m = 0 \rightarrow Q'' = 0, \quad Q(\phi) = Q_0 + Q_1\phi$$

$$\text{Legendre: } P_l(x) = \frac{1}{2^l l!} \left(\frac{d}{dx} \right)^l (x^2 - 1)^l, \quad P_l(1) = 1, \quad \int_{-1}^1 P_l(x) P_l'(x) dx = \delta_{ll'} \left(\frac{2}{2l+1} \right)$$

$$\partial_\phi \Phi(r, \theta, \phi) = 0 \rightarrow \Phi(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos \theta)$$

$$P_l^m(x) = \frac{(-1)^m}{2^l l!} (1-x^2)^{m/2} \frac{d^{l+m}}{dx^{l+m}} (x^2 - 1)^l, \quad \int_{-1}^1 P_l^m(x) P_{l'}^m(x) dx = \frac{2}{2l+1} \frac{(l-m)!}{(l+m)!} \delta_{ll'}$$

$$\text{Spherical harmonics: } Y_{lm} = \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) e^{im\phi}$$

$$Y_{l,-m}(\theta, \phi) = (-1)^m Y_{lm}^*(\theta, \phi), \quad Y_{l0}(\theta, \phi) = \sqrt{\frac{2l+1}{4\pi}} P_l(\cos \theta)$$

$$\int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta Y_{l'm'}^*(\theta, \phi) Y_{lm}(\theta, \phi) = \delta_{ll'} \delta_{m'm}$$

$$\sum_{l=0}^{\infty} \sum_{m=-l}^l Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) = \delta(\phi - \phi') \delta(\cos \theta - \cos \theta')$$

$$\Phi(r, \theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l [A_{lm} r^l + B_{lm} r^{-(l+1)}] Y_{lm}(\theta, \phi)$$

$$\partial_z V = 0 \rightarrow V(s, \phi) = S(s) \Phi(\phi) \rightarrow A_0 + B_0 \ln s + \sum_{n=-\infty, \neq 0}^{\infty} s^n (A_n \cos n\phi + B_n \sin n\phi)$$

cylindrical: $\Phi(\rho, \phi, z) = R(\rho) Q(\phi) Z(z)$, $\frac{1}{\Phi} \nabla^2 \Phi = 0 = \frac{(\rho R')'}{\rho R} + \frac{1}{\rho^2} \frac{Q''}{Q} + \frac{Z''}{Z} = \frac{1}{R} \left(R'' + \frac{R'}{\rho} \right) \pm \frac{\nu^2}{\rho^2} \pm k^2$

$$\frac{Q''}{Q} = -\nu^2 \rightarrow Q(\phi) = e^{\pm i\nu\phi}, \sin / \cos \nu\phi; \quad \Phi(\phi = 2\pi) = \Phi(\phi = 0) \rightarrow \nu = m \text{ (integer)}$$

$$\frac{Z''}{Z} = +k^2 \rightarrow R(\rho) = J_\nu(k\rho), N_\nu(k\rho); \quad \frac{Z''}{Z} = -k^2 \rightarrow R(\rho) = I_\nu(k\rho), K_\nu(k\rho)$$

$$\nu = 0 \rightarrow Q'' = 0, \quad Q(\phi) = Q_0 + Q_1\phi; \quad k = 0 \rightarrow Z'' = 0, \quad Z(z) = Z_0 + Z_1 z$$

$$\nu = k = 0 \rightarrow (\rho R')' = 0, \quad R(\rho) = R_0 + R_1 \ln \rho$$

$$\text{Bessel: } J_\nu(x) = \left(\frac{x}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{x}{2}\right)^{2k}, \quad N_\nu(x) = \frac{J_\nu(x) \cos \nu\pi - J_{-\nu}(x)}{\sin \nu\pi}$$

$$H_\nu^{(1)}(x) = J_\nu(x) + iN_\nu(x), \quad H_\nu^{(2)}(x) = J_\nu(x) - iN_\nu(x), \quad I_\nu(x) = i^{-\nu} J_\nu(ix), \quad K_\nu(x) = \frac{\pi}{2} i^{\nu+1} H_\nu^{(1)}(ix)$$

$$x \rightarrow 0 : J_\nu(x), I_\nu(x) \rightarrow \frac{1}{\Gamma(\nu + 1)} \left(\frac{x}{2}\right)^\nu; \quad N_\nu(x), K_\nu(x) \rightarrow \pm\infty$$

$$x \rightarrow \infty : J_\nu(x)/I_\nu(x) \rightarrow \sqrt{\frac{2}{\pi x}} \cos / \sin \left(x - \frac{\nu\pi}{2} - \frac{\pi}{4}\right), \quad I_\nu(x) \rightarrow \infty, \quad K_\nu(x) \rightarrow \sqrt{\frac{\pi}{2x}} e^{-x}$$

$$\int_0^a J_\nu\left(x_{\nu n} \frac{\rho}{a}\right) J_\nu\left(x_{\nu n'} \frac{\rho}{a}\right) \rho d\rho = \frac{a^2}{2} [J_{\nu+1}(x_{\nu n})]^2 \delta_{nn'}, \quad J_\nu(x_{\nu n}) = 0$$

$$\int_0^\infty J_\nu(k\rho) J_\nu(k'\rho) \rho d\rho = \frac{1}{k} \delta(k - k'), \quad \int_0^\infty J_\nu(k\rho) J_\nu(k\rho') k dk = \frac{1}{\rho} \delta(\rho - \rho')$$

$$\Phi(\rho = a) = 0 \text{ at compact boundary} \rightarrow \text{discrete } k_{mn} = \frac{x_{mn}}{a}, \quad \Phi \propto \sum_{mn} A_{mn} J_m(k_{mn}\rho) Z(k_{mn}z) Q_m(\phi)$$

$$\Phi(z = 0, L) = 0 \text{ at compact boundaries} \rightarrow \text{discrete } k_n = \frac{n\pi}{L}, \quad \Phi \propto \sum_{mn} A_{mn} J_m(k_n\rho) Z(k_n z) Q_m(\phi)$$

$$\text{open radial boundary} \rightarrow \text{continuous } k, \quad \Phi \propto \sum_m \int_0^\infty dk A_m(k) J_m(k\rho) Z(kz) Q(m\phi)$$

Multipole Expansion

$$\rho(|\vec{x}'| > R) = 0, \quad \Phi(|\vec{x}| > R) = \frac{1}{\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{q_{lm}}{r^{l+1}} Y_{lm}(\theta, \phi) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{x}') d^3x'}{|\vec{x} - \vec{x}'|}$$

$$\frac{1}{|\vec{x} - \vec{x}'|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^m \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) \rightarrow q_{lm} = \int_{r' < R} Y_{lm}^*(\theta', \phi') (r')^l \rho(\vec{x}') d^3x'$$

$$\text{Interaction with external } \vec{E} : W = q\Phi(\vec{0}) - \vec{p} \cdot \vec{E}(\vec{0}) - \frac{1}{6} \sum_{ij} Q_{ij} \partial_i E_j(\vec{0}) + \dots$$

Dielectric Media

$$\text{Induced dipole moment density } \vec{P} \rightarrow \vec{\nabla} \cdot \vec{E} = \frac{1}{\epsilon_0} (\rho - \vec{\nabla} \cdot \vec{P}), \quad \vec{D} = \epsilon_0 \vec{E} + \vec{P} \rightarrow \vec{\nabla} \cdot \vec{D} = \rho$$

$$\text{Linear isotropic medium: } \vec{P} = \chi_e \epsilon_0 \vec{E}, \quad \vec{D} = \epsilon \vec{E}, \quad \epsilon = (1 + \chi_e) \epsilon_0, \quad \text{uniform} \rightarrow \vec{\nabla} \cdot \vec{E} = \rho/\epsilon$$

$$\text{Dielectric interface: } (\vec{D}_2 - \vec{D}_1) \cdot \hat{n}_{21} = \sigma, \quad (\vec{E}_2 - \vec{E}_1) \times \hat{n}_{21} = 0, \quad \hat{n}_{21} (1 \mapsto 2)$$

$$\text{Energy: } W = \frac{1}{2} \int \vec{E} \cdot \vec{D} d^3x$$

$$\text{Fourier: } s_m(x) = \sin\left(m\frac{2\pi}{a}x\right), \quad m > 0; \quad c_m(x) = \cos\left(m\frac{2\pi}{a}x\right), \quad m \geq 0$$

$$\int_{-a/2}^{a/2} s_n(x) c_m(x) dx = 0, \quad \int_{-a/2}^{a/2} s_n(x) s_m(x) dx = \int_{-a/2}^{a/2} c_n(x) c_m(x) dx = \frac{a}{2} \delta_{mn}, \quad mn > 0$$

$$f(x) = \frac{1}{2}A_0 + \sum_{m=1}^{\infty} [A_m c_m(x) + B_m s_m(x)], \quad A_m = \int_{-a/2}^{a/2} c_m(x) f(x) dx, \quad B_m = \int_{-a/2}^{a/2} s_m(x) f(x) dx$$

Legendre polynomials

$$P_0 = 1$$

$$P_1 = x$$

$$P_2 = (3x^2 - 1)/2$$

$$P_3 = (5x^3 - 3x)/2$$

$$P_4 = (35x^4 - 30x^2 + 3)/8$$

$$P_5 = (63x^5 - 70x^3 + 15x)/8$$

$$P_6 = (231x^6 - 315x^4 + 105x^2 - 5)/16$$

$$P_7 = (429x^7 - 693x^5 + 315x^3 - 35x)/16$$

Spherical harmonics

$$Y_{00} = \frac{1}{\sqrt{4\pi}}$$

$$Y_{10} = \sqrt{\frac{3}{4\pi}} \cos \theta$$

$$Y_{11} = -\sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi}$$

$$Y_{20} = \sqrt{\frac{5}{4\pi}} \left(\frac{3}{2} \cos^2 \theta - \frac{1}{2} \right)$$

$$Y_{21} = -\sqrt{\frac{15}{8\pi}} \sin \theta \cos \theta e^{i\phi}$$

$$Y_{22} = \frac{1}{4} \sqrt{\frac{15}{2\pi}} \sin^2 \theta e^{i2\phi}$$

$$Y_{30} = \sqrt{\frac{7}{4\pi}} \left(\frac{5}{2} \cos^3 \theta - \frac{3}{2} \cos \theta \right)$$

$$Y_{31} = -\frac{1}{4} \sqrt{\frac{21}{4\pi}} \sin \theta (5 \cos^2 \theta - 1) e^{i\phi}$$

$$Y_{32} = \frac{1}{4} \sqrt{\frac{105}{2\pi}} \sin^2 \theta \cos \theta e^{i2\phi}$$

$$Y_{33} = -\frac{1}{4} \sqrt{\frac{35}{4\pi}} \sin^3 \theta e^{i3\phi}$$